PhysAgent: Physics-Informed SDevice Code Generation via Ontology-Guided LLM

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***Abstract*—As semiconductor scaling reaches its limits, Technology Computer-Aided Design (TCAD) simulation becomes essential for Design-Technology Co-Optimization (DTCO). However, the core challenge lies in configuring SDevice—the physics-and-solver specification stage of Sentaurus TCAD—rather than in device structure generation. We propose PhysAgent, which builds a Sentaurus TCAD guide-based ontology and integrates literature-driven domain knowledge to generate physically grounded SDevice files without relying on synthetic training data.**

***Keywords*—TCAD, ontology, physics-informed modeling, parameter file generation, domain knowledge, semiconductor materials**

I. Introduction

Recent studies have applied LLMs to automate TCAD code generation, primarily targeting structure files written for SDE (Sentaurus Structure Editor). These approaches rely on supervised fine-tuning (SFT) with synthetic datasets constructed by varying numerical parameters in a small number of baseline scripts [3]–[6]. However, this limits generalization to unseen device architectures or materials. More critically, none adequately addresses physics model selection or parameter file generation for SDevice—the core of physically meaningful TCAD simulation. Even AgenticTCAD [6], which generates SDevice code, uses only 2–3 fixed physics configurations and omits parameter files entirely.

We propose PhysAgent, which constructs a Sentaurus TCAD guide-based ontology encoding physical models, equations, and their interdependencies, complemented by literature-driven material-specific knowledge. By providing this structured context to general-purpose LLMs, PhysAgent generates physically grounded SDevice files with explicit reasoning about model selection and parameter settings—without domain-specific fine-tuning.

II. Related Work

TCAD simulation consists of SDE for device geometry and SDevice for physics models and solver settings. While LLMs have been applied to SDE code generation through fine-tuning or in-context learning, SDevice configuration—selecting and parameterizing models such as mobility, recombination, and bandgap narrowing—remains unaddressed due to its high expertise requirements.

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III. Proposed Framework

PhysAgent combines a TCAD physics ontology and literature-driven domain knowledge to guide LLM-based SDevice file generation. The constructed ontology is publicly available as an interactive web platform [7].

A. TCAD Physics Ontology

An SDevice simulation takes a command file (`.cmd`) and a parameter file (`.par`) as input. The physics block in the `.cmd` file specifies the physical models activated for each material, while the `.par` file defines the parameters required by these models along with fundamental properties such as bandgap, permittivity, and the DOS effective masses of electrons and holes. Thus, the model selection in the `.cmd` file determines the parameter set that must be provided in the `.par` file. To consistently capture this `.cmd`–`.par` dependency, we structure the relevant parameters, TCAD syntax, and physical knowledge into an ontology grounded in the governing equations of the physical models and the Poisson and transport equations. Each parameter is represented as a node, and nodes associated with the same physical model are grouped into an equation group. Node–equation and equation–equation relationships are encoded as edges and relations, from which the required parameter set is derived upon model selection. A syntax layer then applies the SDevice syntax to the selected equations and nodes to generate the `.cmd` and `.par` files (Fig. 1).

B. Ontology-Guided Generation

Given a target device and material, PhysAgent accesses the ontology through an MCP (Model Context Protocol) interface that exposes four tools. `ontology_overview` returns the node/edge map of the graph, serving as the entry point for grasping the overall structure. `get_chapter` identifies the equation-level relations associated with a selected node and derives the parameter set required by the corresponding physical equations. `get_reference` returns values and source citations for the derived parameters, drawn from a curated collection of papers, material datasheets, and TCAD references. Finally, `build_syntax` assembles the selected equations and parameter values under the SDevice syntax to generate the final `.cmd` (Physics block) and `.par` files. In our experiments, we employ Claude Sonnet, whose context

window is limited to 200k tokens, whereas the full ontology exceeds 400k tokens and therefore cannot be injected directly into the prompt. The MCP-based selective retrieval scheme circumvents this limitation by allowing the LLM to query only the information required at each step, keeping the per-query context well within the 200k-token budget. Moreover, each parameter in the generated `.par` inherits the citation returned by `get_reference`, thereby enabling parameter-level provenance (Fig. 2).

C. Ontology Expansion and Validation

Unlike SFT-based approaches requiring retraining for new models or materials, PhysAgent uses a directional synchronization workflow with three artifact layers—Spec, Plan, and Impl—where changes propagate automatically from the source of truth while reverse modifications require explicit approval, ensuring incremental expansion without retraining.

IV. Experimental Validations of PhysAgent

PhysAgent targets the physics specification stage—the SDevice `Physics` block and `.par` file—while structure (`.tdr`), solver, and bias schedule remain fixed, orthogonal to prior structure-generation work [3], [5]. Three generators share the same base LLM and prompt: *Vanilla* (spec only), *LLM+Guide* (spec + Sentaurus manual), and *PhysAgent* (spec + ontology + literature). We validate three claims.

Successful TCAD Execution. On a Si nMOSFET we compare three modes (Vanilla, LLM+Guide, PhysAgent) across three prompt levels (L1–L3: explicit model list, categories, intent only), three trials each. Vanilla and LLM+Guide produce characteristic syntactic errors—invented model names, undefined parameters, malformed material blocks—lacking any structural anchor, whereas PhysAgent copies directly from typed ontology nodes at 48× smaller prefill than LLM+Guide. Across L1→L3 model coverage tracks prompt specificity: which physical mechanisms the deck selects depends on what the prompt asserts, making phenomenon-level prompting central to the simulation's analytical scope (Fig. 3).

Literature-Grounded Parameter File Generation.

Phenomenon-Driven Model Selection and Deck Construction.

V. Conclusion

We presented PhysAgent, which generates physically grounded SDevice files by combining a TCAD guide-based ontology with literature-driven domain knowledge, enabling general-purpose LLMs to support expert-level TCAD modeling without synthetic training data.

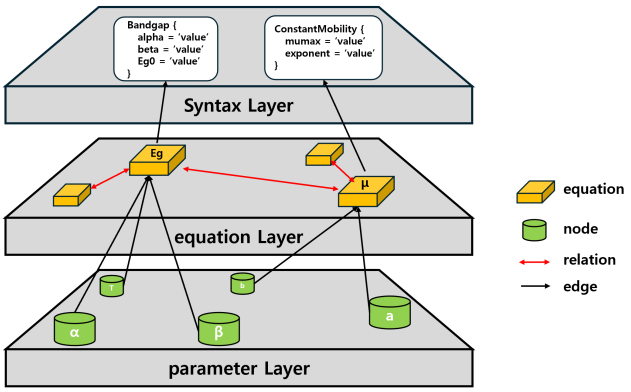

Fig. 1. TCAD physics ontology: node and edge types capturing inter-model dependencies.

Fig. 2. PhysAgent framework: LLM agents access the ontology graph via MCP tool calls to generate physically grounded parameter files.

[Placeholder: Mode × prompt-level comparison on Si nMOSFET—syntactic failure modes and coverage dependence on prompt specificity]

Fig. 3. Successful TCAD execution: mode and prompt-level comparison on Si nMOSFET.

[Placeholder: Model selection accuracy and parameter match rate of PhysAgent vs Sentaurus library across 34 semiconductor materials]

Fig. 4. Reproduction of the Sentaurus library: PhysAgent vs library across 34 materials.

[Placeholder: Id–Vg overlay of PhysAgent-generated a-IGZO TFT vs digitized published measurement]

Fig. 5. Extension beyond the library: a-IGZO TFT Id–Vg from PhysAgent-generated deck vs published measurement.

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